

Pseudocode:

```
// Input:

    argv[1] = Input file name containing employee records.
    Option for user selection:
        • 1 for searching by employee name.
        • 2 for writing records in reverse order to the output file and
          exiting.
// Output:

    • Employee records displayed.
    • Search result for a specific employee name.
    • All records written to output.csv in reverse order.

// High-Level Algorithm:

BEGIN MAIN

// Function Prototypes
DECLARE FUNCTION readFile(filename, records, &line_count)
DECLARE FUNCTION writeFileInReverse(filename, records, count)
DECLARE FUNCTION displayRecords(records, count)
DECLARE FUNCTION searchEmployeeByName(records, count)

// Constants and Variables
CONSTANT MAX_RECORDS = 100
DECLARE records[MAX_RECORDS] of type Employee
DECLARE line_count, choice as INTEGER

// Validate Command-Line Arguments
IF argc != 2 THEN
    PRINT "Usage: <program_name> <input_file>"
    RETURN 1
END IF

// Read Records from Input File
CALL readFile(argv[1], records, &line_count)
IF line_count == -1 THEN
    RETURN 1
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END IF

// Display All Records
CALL displayRecords(records, line_count)

// User Interaction (Options)
PRINT "1. Search by Name"
PRINT "2. Write Data in Reverse to File and Exit"
choice = READ_INTEGER("Choose an option: ")

// Process User Choice
IF choice == 1 THEN
    CALL searchEmployeeByName(records, line_count)
ELSE IF choice == 2 THEN
    CALL writeFileInReverse("output.csv", records, line_count)
    PRINT "Records written in reverse order to output.csv"
ELSE
    PRINT "Invalid option. Try again."
END IF

END MAIN

// High-Level Algorithm End

// Low-Level Algorithm: Start

Step 1 Refinement
FUNCTION readFile(filename, records, &line_count)
    OPEN file with filename in read mode
    IF file is NULL THEN
        PRINT "Error opening file: ", filename
        RETURN -1
    END IF

    SET line_count = 0
    WHILE line_count < MAX_RECORDS AND READ name, age, wage FROM file INTO
records[line_count]

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        line_count = line_count + 1
    END WHILE

    CLOSE file
    RETURN 0
END FUNCTION

```

Step 2 Refinement

```

FUNCTION writeFileInReverse(filename, records, count)
    OPEN file with filename in write mode
    IF file is NULL THEN
        PRINT "Error opening file: ", filename
        RETURN
    END IF

    FOR i FROM count - 1 TO 0 DO
        WRITE records[i].name, records[i].age, records[i].wage TO file in
        CSV format
    END FOR

    CLOSE file
END FUNCTION

```

Step 3 Refinement

```

FUNCTION displayRecords(records, count)
    PRINT "Employee Records:"
    FOR i FROM 0 TO count - 1 DO
        PRINT "Name:", records[i].name, "Age:", records[i].age, "Wage:",
        records[i].wage
    END FOR
END FUNCTION

```

Step 4 Refinement

```

FUNCTION searchEmployeeByName(records, count)
    DECLARE searchName as STRING
    DECLARE found as BOOLEAN = FALSE

    searchName = READ_STRING("Enter name to search: ")

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```
FOR i FROM 0 TO count - 1 DO
    IF records[i].name == searchName THEN
        PRINT "Name:", records[i].name, "Age:", records[i].age,
"Wage:", records[i].wage
        SET found = TRUE
        BREAK
    END IF
END FOR

IF NOT found THEN
    PRINT "Name:", searchName, "not found."
END IF
END FUNCTION

// Low-Level Algorithm: End
```